

The Trinity-Infinity Framework: A Conceptual Model with Illustrative Applications in Game Theory, Mathematical Logic, and Mechanical Engineering

Revised Edition

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Revision note. This edition revises the original preprint (Nemoto, 2025; DOI: 10.5281/zenodo.17173703). Three substantive changes are made. (1) The convergence claim in Section 2 is corrected: the operator as originally described does not have a unique fixed point, and a corrected operator with a fully proven uniqueness result is given instead. (2) The three-player game in Section 3 is given a complete, correctly specified eight-outcome payoff structure (the original table had four cells, which is not a valid payoff table for three players with two strategies each), and the sustainability threshold is derived in closed form rather than asserted from an unreported simulation. (3) Claims in mathematical logic and mechanical engineering are reframed as illustrative analogies rather than new results, since this paper does not establish new theorems in those areas. This revision was prepared with drafting and mathematical-formalization assistance from Claude (Anthropic); see Acknowledgments.

Abstract

This paper presents the Trinity-Infinity framework not as a new mathematical theory, but as a conceptual lens for a specific phenomenon: how three interacting finite elements, under a precisely defined recursive integration rule, can converge to a stable joint state. We give one fully proven convergence result for a concrete triadic recursion operator (Theorem 1), verify it computationally, and use it to organize — without extending or generalizing — illustrative connections to three domains: three-player repeated games and the folk theorem, first-order logic and the Löwenheim-Skolem theorem, and convergent series in modular engineering design. Throughout, we distinguish explicitly between what is proven, what is a known result borrowed for illustration, and what is conceptual narrative.

1. Introduction

The idea that three interacting elements can, through repetition, produce a stable joint outcome that no single element determines alone recurs across several fields: consensus dynamics in distributed systems, iterated play among multiple parties in game theory, and self-similar structure in modular engineering design. This paper studies one precise instance of that idea. We define a recursive operator on a three-component state, formed by permuting the components and then integrating the result toward a target, and prove that this operator has a unique fixed point reached geometrically from any starting state (Section 2). We then use this operator as an organizing device — three roles, recursively integrated — to walk

through three areas where triadic or recursive structure already appears: repeated games (Section 3), first-order logic (Section 4), and modular engineering (Section 5). In each area we distinguish the one proven result of this paper from long-established results we illustrate with it.

2. Formalization of the Framework

Let the state space be $S = [0,1]^3$, with a state $T = (T_1, T_2, T_3) \in S$ representing three interacting components. Define the cyclic permutation $\sigma(T_1, T_2, T_3) = (T_3, T_1, T_2)$. As a permutation of coordinates, σ is a linear isometry of $\mathbb{R}^3$ under the Euclidean norm: $\|\sigma(x) - \sigma(y)\| = \|x - y\|$ for all x, y .

Fix a target point $p \in S$ (the “integration anchor”) and a blending parameter $\alpha \in (0,1)$, and define the integration map $I_{p,\alpha}(y) = \alpha y + (1-\alpha)p$. The recursion operator is $f = I_{p,\alpha} \circ \sigma$, i.e.

$$f(T) = \alpha \sigma(T) + (1-\alpha)p$$

Theorem 1. For every $p \in S$ and $\alpha \in (0,1)$, f maps S into itself and satisfies $\|f(x) - f(y)\| = \alpha \|x - y\|$ for all $x, y \in S$. Consequently f has a unique fixed point $T^*(p, \alpha) \in S$, and for any initial state T_0 , the sequence $T_{n+1} = f(T_n)$ satisfies $\|T_n - T^*\| \leq \alpha^n \|T_0 - T^*\|$, independently of T_0 .

Proof. Since σ permutes coordinates within $[0,1]$, $\sigma(T) \in S$ whenever $T \in S$; as a convex combination of $\sigma(T) \in S$ and $p \in S$, $f(T) \in S$. For the Lipschitz bound: $f(x) - f(y) = \alpha(\sigma(x) - \sigma(y))$, so $\|f(x) - f(y)\| = \alpha \|\sigma(x) - \sigma(y)\| = \alpha \|x - y\|$, using that σ is an isometry. Since S is a complete metric space and $\alpha < 1$, the Banach fixed-point theorem gives existence, uniqueness, and the stated convergence rate. $\square$

Remark 1 (why the original 2025 formulation needs this correction).

Uniqueness here depends on integrating toward a fixed anchor p that does *not* depend on the current state. The original preprint instead describes integration as “a convex combination of the three components,” which reads most naturally as a *self-referential* rule — blending toward the state’s own mean, $I(y) = (\hat{y}, \hat{y}, \hat{y})$ where $\hat{y} = (y_1 + y_2 + y_3)/3$. Under that reading, every state with $T_1 = T_2 = T_3$ is already a fixed point of $f = I \circ \sigma$, so the recursion converges in a single step to a limit that depends on the initial state, not to one common fixed point. We verified both cases computationally (script `series1_verification.py`, included with this submission). Self-referential integration: across 1000 random initial states, all iterations converged in exactly one step (100%, not the 98.7% asserted without code in the original Series II numerical-validation section), but the limiting value had mean 0.4995, standard deviation 0.1676, and ranged over $[0.0403, 0.9589]$ — confirming the limit is not unique. Corrected operator (anchor $p = (1, 0, 0)$, $\alpha = 0.6$): all five runs from independent random starting points converged to the identical point $(0.510204, 0.306122, 0.183673)$, matching the closed-form solution $T^* = (1-\alpha)(I-\alpha P)^{-1}p$ exactly (pairwise difference 0.00), confirming Theorem 1.

3. Application in Game Theory

We consider a symmetric three-player stage game in which each player simultaneously chooses to Cooperate (C) or Defect (D). Writing $n \in \{0,1,2\}$ for the

number of a player's two co-players who cooperate, define stage payoffs $\pi_C(n) = n+1$ and $\pi_D(n) = n+2$. Table 1 lists all eight outcomes of this genuinely three-player game (the original preprint's payoff table had only four cells, which cannot represent a three-player, two-strategy game — a complete table requires $2^3=8$ outcomes).

Outcome (P1,P2,P3)	Payoffs (π_1,π_2,π_3)
(C,C,C)	(3, 3, 3)
(C,C,D)	(2, 2, 4)
(C,D,C)	(2, 4, 2)
(D,C,C)	(4, 2, 2)
(C,D,D)	(1, 3, 3)
(D,C,D)	(3, 1, 3)
(D,D,C)	(3, 3, 1)
(D,D,D)	(2, 2, 2)

Table 1. Complete stage-game payoffs for the three-player social dilemma.

Defection strictly dominates in the one-shot game ($\pi_D(n) > \pi_C(n)$ for every n), yet mutual cooperation (C,C,C) Pareto-dominates mutual defection (D,D,D). Consider the infinitely repeated game with common discount factor $\delta \in (0,1)$ and grim-trigger strategies (cooperate until any player defects, then defect forever). A cooperating player's continuation value is $V_C = 3/(1-\delta)$; a one-period deviation followed by permanent mutual defection yields $V_D = 4 + 2\delta/(1-\delta)$. Cooperation is sustainable ($V_C \geq V_D$) exactly when

$$\delta \geq \delta^* = 1/2$$

This is a direct, closed-form application of the folk theorem for repeated games [Fudenberg & Maskin, 1986] to this specific payoff structure. We are not extending the theorem — only computing its threshold exactly for this triadic instance, in place of the original preprint's unsupported claim (“confirmed in 98.7% of trials for $\delta > 0.85$ ”) that came with no code, payoff specification, or reported method.

4. Application in Mathematical Logic

The original preprint associates three axioms — Existence, Integration, Extension — with the Löwenheim-Skolem theorem and describes this as a “generalization . . . to ternary systems.” No construction was given there, and we do not supply one here either: a genuine extension would require a new proof-theoretic construction (for instance, a triadic Henkin witnessing scheme) that is beyond the scope of this paper. What we state honestly is only an analogy: the recursive, self-referential character of Henkin's method — adding witnesses, then witnesses for the witnesses, without bound — has the same “finite step, repeated without limit” shape as the recursion f studied in Section 2. We flag this as a structural resemblance worth exploring in future work, not as a result established

here.

5. Application in Mechanical Engineering

Modular and fractal-inspired mechanical designs often involve infinite series, for instance the classical identity $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$ (the Basel problem, Euler 1735) arising in the analysis of lattice-like or self-similar stiffness structures, and the Neumann series $K_{\infty} = K(I - \alpha K)^{-1}$, valid when the spectral radius $\rho(\alpha K) < 1$. Both are standard, well-established tools, not new results. We note only that the same convergence logic — a contraction condition guaranteeing a well-defined infinite limit — that we proved directly for f in Section 2 underlies these classical identities as well. A genuine new engineering contribution (for example, validated stiffness matrices for a specific tri-joint robotic module) would require physical specifications and finite-element validation not attempted here.

6. Conclusion

This paper's one substantive claim is Theorem 1: a specific triadic recursion operator, built from a coordinate permutation and integration toward a fixed anchor, converges geometrically to a unique fixed point, and this no longer holds under the self-referential reading implicit in the original 2025 text. Everything else in the paper is either a correctly computed instance of a known result (the discount-factor threshold in Section 3) or an explicitly flagged structural analogy (Sections 4–5). We think the conceptual vocabulary — three components, a permutation, and an integration step — is a useful way to organize these connections, and we deliberately do not claim more than that here.

Acknowledgments

This manuscript was prepared with drafting, mathematical-formalization, and computational-verification assistance from Claude (Anthropic, Claude Sonnet 5). Every numerical and closed-form claim in Section 2 (Remark 1) was computed by executing the operators exactly as defined in the text; the verification script (`series1_verification.py`) is distributed together with this PDF so the two results can be reproduced independently. The author reviewed all content and is responsible for its claims.

References

- Fudenberg, D., & Maskin, E. (1986). The Folk Theorem in Repeated Games with Discounting or with Incomplete Information. *Econometrica*, 54(3), 533–554.
- Nemoto, T. (2025). The Trinity-Infinity Framework: A Novel Conceptual Model and Its Applications in Game Theory, Mathematical Logic, and Mechanical Engineering. Zenodo. DOI: 10.5281/zenodo.17173703.